Lab 4: VibeCoding — LLM Chat History

Student: Merull Shah (PES1UG24CS270, Section E) | Repo: SETAPESU26/25_zombie_escape | Tool: Claude (Opus 5.5) on claude.ai | Date: 11 Oct 2026

Prompt 1 — User

[Attached: Lab_4_VibeCoding_Student_handout.pdf]
https://github.com/SETAPESU26/25_zombie_escape

Response 1 — Claude

I'll clone the repo, read the README for the bugs and feature list, and get the game running first.

■ Cloned the repo; read README.md (Tasks 1–4: Health System, Ammo System, Explosive Barrel, Zombie Types) and game.py (199 lines). Installed pygame-ce 2.5.8.

Let me confirm the bugs I spotted by running the original headlessly.

■ Ran a probe script against the original code. Output:

```
score after kill (expect >=10): 10        <- overwritten next frame by timer
zombies left, kills, need: 0 4 8           <- no zombies left, wave can never finish
zombie moved in 60 frames toward (100,3) dir: 60 0 (expect ~90, ~2.7)
zombie moving up-left moved: -60 -60
HUD width px: 938 screen 800
```

Confirmed four bugs in the original before touching anything:

• **Wave progression stalls** — wave 1 spawns 4 zombies but needs 8 kills, and new zombies only spawn on wave change. Kill all 4 and the game sits empty forever.

• **Zombie movement truncated** — `int()` on a 1.5 speed means zombies actually move at 1 px/frame, and any small direction component rounds to 0.

• **Score is wiped every frame** — `self.score = int(time…)` overwrites the +10 kill bonus on the next tick.

• **HUD overflows** — 938 px of text on an 800 px window.

■ Rewrote game.py: float positions for zombies; continuous trickle spawning per wave; score = seconds + kill points; HUD with hearts/ammo; Player.hp + invincible_until; Player.ammo + reload_until; Barrel/Explosion classes + place_barrels(); FastZombie/TankZombie subclasses + weighted pick_zombie_type(wave).

■ Ran a test script against every README "Done when" criterion with a simulated clock. Result:

```
PASS 1 touch -> hp 2, alive
PASS invincible during window (still touching)
PASS 2nd hit after window -> hp 1
PASS game over only at 0 hp
PASS restart restores hp
PASS 12 shots empties clip, reload starts
PASS shooting blocked at 0
PASS not restored at 1.9s
PASS reload countdown correct (0.1s left)
PASS restored at 2.0s
PASS barrel removed on bullet hit
PASS zombies in radius destroyed
PASS zombie outside radius survives
PASS explosion effect active
PASS barrel kills counted
PASS 4 barrels placed
PASS fast: 1hp, faster, smaller
PASS tank: 6hp, slower, larger
PASS tank takes exactly 6 hits
  wave3 mix: {'Zombie': 1170, 'TankZombie': 312, 'FastZombie': 518}
PASS waves keep progressing (reached wave 15)
PASS kill bonus kept in score
PASS zombie moves full speed at shallow angle (90,5)
```

■ Rendered a frame to check the HUD and visuals; delivered game.py.

Summary given: Bugs fixed — waves no longer stall; zombies move correctly; kill bonus stays in the score; HUD fits the window. Features — (1) Health: 3 hearts, 1.5 s invincibility with blink + white outline, "DEVOURED!" only at 0 HP. (2) Ammo: 12-round clip, auto 2 s reload with on-screen countdown, shooting blocked until complete. (3) Barrels: 4 barrels, 120 px blast kills zombies in radius, kills count toward

score/wave, barrel removed. (4) Zombie types: Fast (small, 2× speed, 1 HP) and Tank (large, ~half speed, 6 HP, health bar), mixed into waves and more common later. Advice: record the 'before' video from a fresh clone first; commit to Lab-4/ in own repo; don't open a PR to SETAPESU26.

Prompt 2 — User

I need all my deliverables

Response 2 — Claude

I'll produce all three: before/after gameplay videos rendered from the actual code, the updated code, and the chat history as a PDF, bundled in a Lab-4 folder.

- ■ Wrote a recorder that drives the game with scripted keyboard/mouse input under a simulated clock and pipes 600 frames (10 s @ 60 fps) to ffmpeg.
- ■ before.mp4 (original code): the bot kills all 4 starting zombies, then the map stays empty at Wave 1 — the stall bug on camera.
- ■ after.mp4 (fixed code): first attempts had the barrel shot land on a cooldown/reload frame and bullets absorbed by zombies in front of the barrel; restaged the scene with a clear line of fire, retried the barrel shot until it connects, and added dodging so the clip ends with the player alive. Final video shows: barrel blast wiping a pack (kills 1→5), a hit costing one heart with the invincibility outline, reload countdown, Fast/Tank zombies with health bars, and progression to Wave 2.
- ■ Generated this chat-history PDF and packaged Lab-4/ (before.mp4, after.mp4, game.py, chat_history.pdf).